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Design and implementation of a web-based and open-source hero administration for the role-playing game “Das Schwarze Auge” (Backend)

Bachelor thesis
Final presentation

Björn Nowak, 25.05.2023

1 Introduction
1.1 Motivation

Pen **&** **P**aper
Role-**P**laying **G**ame

“Das Schwarze Auge” (DSA) germanys most popular P&P RPG

The problems:



Time-intensive and error-prone character creation and management for players

- A complex rulebook
  - Vast amount of rules
  - Different categories of rules for different game phases
  - Countless combinations of character feature forms
- Many rulebook extensions increasing complexity further

1 Introduction
1.1 Motivation

“Das Schwarze Auge” (DSA) germanys most popular P&P RPG

The countermeasures:



Various tools supporting players keeping an overview and automating tasks

- E.g. character layouts, spread sheets and even programs
- Mainly (almost exclusively) created and published by the community

Current restriction <u>which this work tackles</u>:



- Non of the popular programs supports the current edition
- Reportingly due to required extensive work and lack of developing capacities

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Outline

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- 3.3 Rulebooks logic programs (LPs)
- 3.4 Encountered difficulties

4 Discussion and outlook

5 Conclusion

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1 Introduction

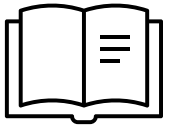
~~1.1 Motivation~~

1.2 Objectives

1.3 Basics of “Das Schwarze Auge” (DSA)

1 Introduction
1.2 Objectives

Creating a cookbook for formalizing rules of the current (5th) DSA edition, that…

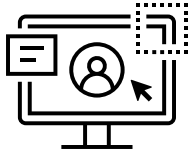


– provides general conventions

– defines a character model

– outlines pre-conditions for processing

Creating a web-based and open-source software as a platform, which…



– implements above cookbook as a proof-of-concept

– enables easy migration to other editions

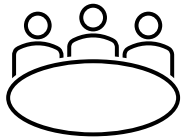
– supports multiple rulebook extensions

– is very maintainable

Overall focusing on base rules at the character creation/validation

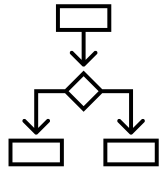


1 Introduction
1.3 Basics of “Das Schwarze Auge” (DSA)



Involved roles:

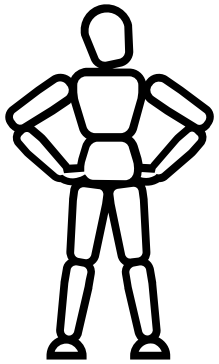
- One game master
- Characters also called Heros (remaining participants/players)



Game process:

- Participants create their character
- The game master leads the players through a fictional world using a pre-defined rulebook (interactive story-telling)
- Characters may gain adventure points (AP) during a session and use this to advance themselves
- Eat, sleep and repeat

1 Introduction
1.3 Basics of “Das Schwarze Auge” (DSA)



of four
of 25
of 33
of eight
of 160
of 222
of 50
of 52

- 59 talents
- 45 spells
- six rituals
- 16 cantrips
- 34 liturgies

e.g. **not** including combat techniques

$$\binom{R}{1} * \binom{C}{1} * \binom{P}{2} * \binom{CH}{6} * \binom{\frac{S}{4} + \frac{SP}{10}}{20} * \binom{\frac{A}{2}}{10} * \binom{\frac{D}{2}}{5} = \text{approx. } 3.16078 \text{ decillion } (10^{33}) \text{ combinations}$$

(equation includes assumptions)

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2 Design

2.1 Used formalization method

2.2 Concept

2.2.1 Conventions

2.2.2 Input model

2.2.3 Preconditions

2.2.4 Cookbook for rule formalization

2.2.5 Result interpretation

2 Design
2.1 Used formalization method

Answer Set Programing (ASP)	
Principal	Finds solutions non-deterministic under constraints / rules
Language	Normal-form horn-clauses
Technique	Stable Model Semantics (Answer Set), Grounder, Solver
Application	Constraint-based modelling, Combinatorial optimization, planning problems and decision support
Commonness	Wider than Prolog and SAT *
Strengths	Precise and complete representation of knowledge, support of negation and uncertainty, complex program flow control
Weaknesses	Complex syntax and semantic hence steep learning curve

* Based on Google Scholar hits using the name and abbreviation

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Used framework:



clingo
from Potassco (the Potsdam Answer Set Solving Collection)

Used programing language:



python

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2.2 Concept / 2.2.1 Conventions

2 Design
2.2 Concept / 2.2.2 Input model

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2.2 Concept / 2.2.3 Preconditions



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3 Implementation

3.1 General process
3.2 Core components
3.3 Rulebooks logic programs (LPs)
3.4 Encountered difficulties

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3 Implementation
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4 Discussion and outlook

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4 Discussion and outlook
A pre-discussion

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4 Discussion and outlook
An outlook

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5 Conclusion

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5 Conclusion

– Soon available for contributions at: https://github.com/bjoern-nowak/dsa-heldenbogen